Jun Young Kim

Ph.D. Student, North Carolina State University

+1 (919) 345-9804 jkim249@ncsu.edu linkedin/lgkimjy github/lgkimjy lgkimjy.github.io

Education

North Carolina State University Aug. 2025 – present
Ph.D., Department of Mechanical and Aerospace Engineering *NC, United States*

Hanyang University Mar. 2014 – Aug. 2022
Master of Science, Department of Interdisciplinary Robot Engineering Systems *Ansan, Republic of Korea*
Bachelor of Science, Department of Robotics

Research Interests

Humanoid Loco-Manipulation	Hybrid Learning and Control	Whole-Body Control
Safety-Critical Control	Dynamic Locomotion	Optimization and RL

Publications

Manuscripts and Preprints

[m2] **J. Y. Kim**, J. Lee. "An Integrated Framework for Safe and Balanced Humanoid Manipulation via Contact-Aware Planning and Whole-Body Control," manuscript submitted to *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[m1] **J. Y. Kim**, J. An, Y. Oh. "Reactive Locomotion Control for a Wheeled-Legged Humanoid Robot Leveraging Task-Prioritized Whole-Body Control," 2025

Journal Articles

[j1] J. An, **J. Y. Kim**, M. T. Lim, Y. Oh. "High-Speed and Enhanced Motion Control for a Wheeled-Legged Humanoid Robot Using a Two-Wheeled Inverted Pendulum With Roll Joint," *IEEE Access*, 2025 [doi]

Conference Papers

[c4] S. H. Lee, D. Noh, **J. Y. Kim**, J. Yong, J. Y. Kim, J. K. Han. "Enhanced Localization for Bipedal Humanoid Robots in Dynamic and Noisy Environments," *2025 10th International Conference on Control and Robotics Engineering (ICCRE)*, 2025 [doi]

[c3] **J. Y. Kim**, M. S. Ahn, J. K. Han. "Enhancing AdultSize Humanoid Localization Accuracy: A Vision-based aMCL Leveraging Object Detection Model and Hungarian Algorithm," *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*, 2023 [doi]

[c2] H. J. Kim, D. K. Oh, **J. Y. Kim**, M. S. Kang. "Development of a Sensor Module for Proximity Communication and Obstacle Detection of Handrail Mounted Type Walking Assists," *Korean Society for Precision Engineering*, 2018

[c1] H. J. Kim, H. J. Kim, D. K. Oh, **J. Y. Kim**, H. J. Byun, M. S. Kang. "Learning to Generate Trajectory of Striking Motion for Two-Link Robot Arm in Air Hockey," *Korean Society for Precision Engineering*, 2017

Research Experience

Graduate Researcher | **Hybrid Intelligent Experimental Robotics (HIER) Lab** Aug. 2025 – present
North Carolina State University (Advisor: Prof. Jaemin Lee) *NC, United States*

- **Hybrid Learning and Model-Based Loco-Manipulation**: Developing humanoid loco-manipulation methods that combine learning-based policies with model-based planning and whole-body control.
- **Kinodynamic Contact Planning**: Proposed an integrated framework that combines feasibility analysis, contact planning, and whole-body control for safe and balanced task execution. [project]

Research Associate | Humanoid Robotics Laboratory Mar. 2023 – Dec. 2024
Korea Institute of Science and Technology (Advisor: Dr. Yonghwan Oh) *Seoul, Republic of Korea*

- **Dynamic Locomotion for Wheeled-Legged Humanoid**: Developed a DCM-based online footstep planner that reactively computes foot placement and step timing. [demo]
- **Hybrid Locomotion for Wheeled-Legged Humanoid**: Unified walking and driving modes using convex MPC to improve high-mobility locomotion. [demo]

Visiting Graduate Researcher | Robotics & Mechanisms Laboratory (RoMeLa) Jan. 2022 – Jul. 2022
University of California, Los Angeles (Advisor: Prof. Dennis Hong) *CA, United States*

- **Humanoid Localization in Soccer Field**: Developed a vision-based localization system for a humanoid robot platform by integrating a particle filter with an object-detection model. [demo]

Graduate Researcher | HERoEHS Laboratory Mar. 2020 – Jan. 2022
Hanyang University (Advisor: Prof. JeaKweon Han) *Ansan, Republic of Korea*

- **Adult-Size Humanoid Robot Development**: (1) Developed a kinematic controller for humanoid robot motion generation and (2) a computer vision-only localization.
- **Robot Systems Integration**: Integrated UWB into balloon-robot systems for multi-robot positioning.
- **Safer Robot Interaction**: Led software and hardware teams to develop a mecanum wheeled robot; designed loadcell-based force response control with a Mass-Spring-Damper model. [demo]
- **Autonomous System for Dual-Arm Mobile Manipulator**: (1) Implemented solid-state LiDAR with LOAM-based algorithm for indoor 3D SLAM. (2) Integrated Speech-to-Text for human-robot interaction.

Awards

NC State University Graduate Fellowship, **Office of the Provost**	2025–2026
RoboCup 2022 Bangkok Humanoid League AdultSize Soccer Competition, **2nd Place**	2022
RoboCup Korea Open 2020 Humanoid AdultSize Soccer, **1st Place**	2020

Teaching and Mentoring

Teaching Assistant, MAE 405: Controls Lab, North Carolina State University	Aug. 2025 – present
Teaching Assistant, Innovation Center for Engineering Education	2020 - 2022
Student Mentor, Gangwon Career Education Institute	Jun. 2019 - Aug. 2019
Education Mentor, Hanyang ERICA Summer School 2019 (HESS2019)	Jul. 2019

Certifications

Nominated, Smooth Sailing Cohort, NC State College of Engineering	2025–2026
H-Mobility Class (Autonomous Driving Course), Hyundai Motor Company	2022
Accreditation Board for Engineering Education of Korea (ABEEK) Completion	2020

Services-Review

IEEE Robotics and Automation Letters (RAL)	2026
IEEE International Conference on Robotics and Automation (ICRA)	2024